From Alerts to Actions: An Action-Centered Human–AI Decision Support Framework for Reliable Disaster Response

Geunsik Lim

***Abstract*—Decision systems involving human and AI are being used in critical situations where inappropriate actions on the part of the human could result in Human–AI decision systems are increasingly used in situations where an inappropriate or delayed human action can have serious consequences. Even though there has been an improvement Improvements in the predictive capabilities of AI , it does notautomatically translate to appropriate actions on the part of the human . The focus of this paper is to understand how the structure of a human–AI decision systemaffects action execution, trust, and cognitive workload in uncertain situations. We introduce a new concept for human–AI decision system design: *Action-Centered Human–AI Decision Framework (ACHDF)*, which focuses on action execution rather than on the performance of the algorithm. Our mixed-method evaluation strategy using controlled user studies accuracy of AI do not, by themselves, translate into appropriate human action. Drawing on the joint cognitive systems perspective, which evaluates the human and the machine as a single system, we present the *Action-Centered Human–AI Decision Framework (ACHDF)*. ACHDF instantiates uncertainty-aware risk estimation, guardrailed AI recommendations, and human-in-the-loop escalation for disaster-response decision support, and it treats human action execution as the primary evaluation outcome. In a counterbalanced within-subjects study ($n = 52$), simulations of past disasters, and a pilot deployment in a municipality empirically shows the efficacy of we compared an **ACHDF-based** human–AI decision support systems in significantly outperforming a dashboard-based system in terms of action execution rates, response times, cognitive workload , trust, while avoiding technology overdependence. Our decision problems are not limited to disaster response but include a broader set of decision problems characterized by high levels of uncertainty, asymmetry in error costs, time pressure, and human accountability. Our work thus generalizes the human–machine system literature on prototype with a conventional dashboard in a simulated disaster-response decision task. Compared with the dashboard, the ACHDF condition was associated with a higher action execution rate (0.81 vs. 0.62), shorter response latency (8.1 vs. 12.4 s), lower self-reported workload (NASA–TLX), and higher perceived trust; usability scores were also higher, although no significance test was reported for them. Only the significance thresholds of the original analysis are available, so effect estimates should be interpreted with caution. Because the two conditions differed in several features at once, the results describe the effect of the integrated ACHDF configuration rather than of individual components. We discuss implications for the design of **human–AI decision support** for reliable action executionunder uncertainty. Validation in other domains and with professional operators remains future work.**

***Index Terms*—Human–AI decision systems,** joint cognitive systems, adaptive automation, **human-in-the-loop, trust** calibrationin automation**, cognitive workload, action execution**

I. Introduction

G. Lim is with the Department of Electrical and Computer Engineering, Sungkyunkwan University, 2066, Seobu-ro, Jangan-gu, Suwon 16419, South Korea (e-mail: leemgs@g.skku.edu).

fact, human–AI

HUMAN–AI decision systems are being increasingly used in situations where there is with high uncertainty, high consequence, and high speed [1]–[3]. These situations stakes, and severe time pressure [1]–[3]. Examples include healthcare triage, supervision of autonomous transportation systemsvehicles, industrial safetysystems, and emergency response systems [4]–[9]. In these situations, artificial intelligence is being used to enhance the ability to predict situations accurately. However, human–machine systems researchhas consistently demonstrated that settings, AI is mainly used to improve prediction. Human–machine systems research, however, has repeatedly shown that better predictions do not always lead to necessarily lead to appropriate human action [10]–[12]. In disaster contexts, effective response further response, effective action also depends on community-level coordination, resilient communications, and distributed sensing and reporting pipelines [13]–[15]. Decision making under conditions of uncertainty is an important aspect of human–machine systems research [1], [3], [9]. When there [13]–[15].

When information is incomplete or uncertaininformation, human operators are required to consider , operators must weigh risk, confidence, and responsibility in situations where they interact with machines . Previous research has demonstrated the fundamental influence of conditions of uncertainty on human action. In the context of while working with machines [1], [3]. AI systems, conditions of uncertainty are often not well communicated or are implicit [16]–[18].

In many systems, the results of an AI's computation are displayed as a set of probabilities or as an alarm, but often without adequate support for how these probabilities should guide however, often communicate their uncertainty poorly or only implicitly [16]–[18]. Their outputs are typically shown as probabilities or alarms, with little support for turning them into action [1], [18]. Ultimately, it The human is left to the human to close the loop between prediction and action, and often in situations of often under cognitive stress [12], [19]. A further important question is how human–AI systems can be designed for decision-making under uncertainty [10]. Machine learning advances have greatly enhanced the accuracy of predictions for a broad range of applications [2], [20]. However, human factors research has also suggested that

Although machine learning has greatly improved predictive accuracy [2], [20], accuracy does not necessarily translate into human performance [3], [10]. For example, humans might Operators may wait for confirmation before acting, or might choose not to follow an AI's advice when it is not trusted, or might become overly reliant disregard advice they do not trust, or over-rely on an AI when its whose accuracy is overstated [12], [21], [22]. This illustrates one of the important shortcomings associated with prediction-focused system design approaches. The

execution of human actions not only depends [12], [21]–[23]. Human action therefore depends not only on what the system predicts, but also on how the prediction is communicated and how the system provides control to control is shared with the user. Without specific tools to address issues like trustcalibrationexplicit support for appropriate trust, workload management, and human override, there is a risk that the system could exacerbate decision support can increase automation bias or cognitive workload , thus negating the benefits of the system [1], [12], [19], [24]. workload and lose its intended benefit [1], [12], [19], [25].

While there has been extensive research on issues like automation , trust, and decision-making, there is a lack of empirical research on how specific human–AI system design decisions influence measurable action execution outcomes. The evaluation of most human–AI systems has been conducted on the basis of their These concerns are not new to cognitive systems engineering. The joint cognitive systems (JCS) perspective holds that a human and a machine working together form a single cognitive system, and that this system should be evaluated by the outcomes it jointly produces, not by the performance of the machine alone [26], [27]. Research on adaptive automation and levels of automation has likewise studied how support should change with task demands while keeping the human in control [28]–[30]. Much current work on AI-based decision support, however, still evaluates systems by prediction accuracy or usability , not on how well the system enables the human to execute appropriate actions under uncertain conditions rather than by whether the human–AI system produces appropriate and timely action [3], [9]–[11]. This paper aims to applies the JCS perspective to AI-assisted disaster-response decision support and tests it empirically.

We address the following, deliberately scoped research question:

> *How can* Compared with *a human–AI* conventional dashboard, is an ACHDF-based *decision* support *system be designed* associated with better action execution in a simulated, high-uncertainty disaster-response task, and how does it relate *to reliably support timely* response latency, workload, *and appropriate human action under conditions of high uncertainty* trust*?*

In order to answer this question, we will examine how uncertainty-aware risk estimation, guardrailed AI recommendations, and human-in-the-loop escalation mechanisms impact action execution The study tested the following hypotheses, which were defined before data collection:

- **H1 (primary):** The Action Execution Rate (AER) is higher in the AI-assisted (ACHDF) condition than in the baseline condition.
- **H2 (secondary):** Response latency is shorter in the AI-assisted condition.
- **H3 (secondary):** Self-reported workload (NASA–TLX) is lower in the AI-assisted condition.
- **H4 (secondary):** Perceived trust and usability are higher in the AI-assisted condition.

In this study, action execution is the primary outcome. Response latency, workload, trust, trust calibration, and cognitive workload [16]–[18], [31]. The main and usability are secondary outcomes. Trust and workload also act as *design considerations* that shaped how information is presented, but the implemented system does not measure or adapt to them at run time (Section III-A).

The contributions of this paper to the research on human–machine systems include the following: are as follows. First, we define *action execution* as an important outcome variable for evaluating apply the JCS principle of evaluating joint human–machine outcomes to AI-assisted decision systems, going beyond accuracy disaster decision support, with action execution as the primary measure for evaluating AI systems [3], [10]evaluation outcome. The principle itself is established in cognitive systems engineering [26], [27]. Our contribution is to make it concrete for an application area where AI systems are usually evaluated by accuracy alone. Second, we introduce a human–AI decision architecture that incorporates uncertainty, constraints on AI recommendationsthrough guardrails, and human-in-the-loop escalation and override capabilities [1], [31]present ACHDF, a domain-specific instantiation of the JCS and adaptive function allocation perspectives that combines uncertainty-aware risk estimation, guardrailed AI recommendations, and uncertainty- and criticality-triggered escalation, while the human keeps final decision authority. Third, we conduct an empirical evaluation of our design and find its benefits for human report a controlled within-subjects evaluation ($n = 52$) that compares the integrated ACHDF configuration with a conventional dashboard on action execution, cognitive workload, and trustcalibration in a high-stakes decision task [1], [2].

In addition to contributing to the body of knowledge in terms of specific results, this study also adds to the overall body of knowledge in human–machine systems by extending previous research in trustcalibration, automation biases, and function allocation. Most notably, this study provides an empirical link between uncertainty-aware function allocation and human action execution in a way that can have significant effects on decision-making in high-stakes situations. This study seeks to address a long-standing gap between human factors theory and the practical design of human–AI systems [3], [10], [12]. response latency, workload, trust, and usability. Because several features differed between conditions, this evaluation does not isolate the contribution of individual ACHDF components.

II. Related Work

In this section, we will discuss previous work that is related to the topic of human–AI decision systems, with a focus on the effects of automated decision support on the decision-making process of humansThis section reviews four research streams that ACHDF builds on: joint cognitive systems and adaptive function allocation, which form the conceptual basis of the framework; human–machine decision support; trust, trust in decision systems, and the execution of behavior in critical domains [1]–[3], [10]. We will structure the previous work around three related research streams.

A. *Information Fusion and Network-Based Situational Awareness*

Many workload, and automation bias; and AI assistance in high-stakes decisions in disaster response are mediated by information ecosystems that combine heterogeneous sensor streams, ad hoc communications, and rapidly evolving social information. Resilience and early-warning systems often rely on wireless sensor networks and distributed reporting, where security and robustness are necessary to preserve actionable signal quality under disruption [15]. Beyond physical sensors, post-disaster response may also incorporate opportunistic proximity-based data collection and community reporting, which can improve resource allocation and coordination under severe infrastructure constraints [13], [14]. Network-analytic methods such as community detection provide one mechanism for structuring these large, noisy information

environments. Prior work has examined scalable and model-based community discovery, resolution limits, and benchmark generation for large graphs [32]–[38]. Related studies have investigated overlapping and fuzzy community structure, which is useful when actors, organizations, and communication channels participate in multiple simultaneous operational groups [39]–[44]. While these graph-mining methods are not a core focus of our decision framework, they motivate how diverse upstream information sources can be structured and summarized so that downstream human–AI decision support can remain action-oriented under time pressure [24]environments. It closes with a short note on the information sources that feed disaster decision support.

A. Joint Cognitive Systems and Adaptive Function Allocation

Cognitive systems engineering treats a human operator and the technology the operator works with as a *joint cognitive system* (JCS). The unit of analysis is the joint system and the outcomes it produces in its work environment, not the machine's internal performance [26], [27]. From this perspective, evaluating decision support by the actions the joint system executes is a natural consequence of the design stance, not a new concept. ACHDF adopts this stance explicitly.

A second relevant body of work addresses how functions should be allocated between humans and automation. Parasuraman *et al.* distinguish types of automation (information acquisition, information analysis, decision selection, and action implementation) and levels of automation within each type [28]. Adaptive automation changes the level or type of support in response to task demands, operator state, or critical events [29], [30]. Empirical studies show that intermediate levels of decision support can improve performance, but that higher levels can reduce situation awareness and leave operators poorly prepared to intervene when automation fails [23], [29].

ACHDF relates to this literature in three ways. (i) In the terms of Parasuraman *et al.* [28], it automates information analysis and decision selection (risk estimation and recommendation), but it never automates action implementation: the human always executes the action. (ii) Its adaptation is *event-triggered*. The form of support changes when estimated uncertainty, action criticality, or a policy check crosses a threshold (Algorithm 1), not in response to measured operator workload or physiological state. (iii) What changes is the *level and form of AI support*, from a bounded recommendation to an escalated presentation of evidence and options. Decision authority does not move between the AI and the human. ACHDF therefore does not propose a new allocation theory. It is a domain-specific instantiation of these principles for AI-generated recommendations in disaster response.

B. Human–Machine Decision Support Systems

The human-machine decision support systems were widely researched in the context of Human–machine decision support has been studied extensively in supervisory control, command and controlsystems, and complex socio-technical systems [1], [3], [12]. The research in this domain initially focused on the potential of automation in supporting human decisions in terms of enhancing the speed and consistency of information processing and the potential dangers Early work focused on how automation can make information processing faster and more consistent, and on the risks of misallocating functions between humans and automation systems [12], [19]. More recent research has highlighted the importance of work shows that what matters is not only the quality of decision support information the information provided, but also the manner in which recommendations are incorporated into human decision-making processes how recommendations enter the human decision process [10], [11]. Systems that are limited to information display or prediction may place the burden of interpretation and decision-making on the human user, especially in time-critical situations [1], [2]. Recent research, therefore, suggests the need for decision support systems that go beyond information display to provide only display information or predictions leave interpretation and decision making to the user, which is especially costly under time pressure [1], [2]. This has motivated decision support that provides actionable recommendations while maintaining human control and responsibility keeping the human in control and responsible [3], [31].

C. Trust, Workload, and Automation Bias

Trust and cognitive workload are important constructs used to explain the performance of human–machine systems [1], [12]. Trust has been used to explain the extent to which users of decision support systems, for example, utilize the support, ignore the support, or misuse the support [3], [10]. Previous research has demonstrated the negative effects of over-trust and under-trust on the performance of decision support systems, with over-trust leading to in automation is commonly defined as the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability. *Appropriate* (calibrated) trust means that reliance matches the actual capabilities of the automation [25]. Over-trust can lead to misuse and automation bias, thereby failing to detect errors , and under-trust leading to the failure to utilize decision support systems, thereby increasing workload [21], [22]in which errors made by the automation go undetected. Under-trust leads to disuse of useful support [21]–[23], [25].

Automation bias has been demonstrated to result when decision support systems provide recommendations but fail to represent their uncertainty [17], [21]. On the other hand, the representation of uncertainty and the opportunity for humans and complacency are most likely when automation is highly but not perfectly reliable and when its uncertainty is not represented [17], [23]. Representing uncertainty and giving humans the opportunity to verify decisions has been demonstrated to increase trust levels [1], [18]. Similarly, the role of workload, as measured by proposed as a way to support appropriate reliance [1], [18]. Assessing calibration directly requires comparing reliance with the correctness of the automation's advice, including trials in which the advice is wrong [23], [25]. Workload is commonly measured with the NASA Task Load Index (NASA–TLX) , has been demonstrated to mediate [45] and has been examined as a factor in the relationship between automation and performance [3], [12]. Decision support systems that reduce information processing that reduces information integration and action selection can be used demands is expected to reduce workload and increase the speed of responses by humans response time [10], [11].

D. AI Assistance in High-Stakes Environments

In high-stakes environments such as healthcare, aviation, and emergency management, AI-assisted decision systems are being increasingly employed to facilitate increasingly used for time-critical decisions [1], [2], [6], [7]. Although machine learning has helped improve the predictive accuracy of AI systems in such environments, empirical studies have demonstrated that the predictive ability of AI in healthcare, aviation, and emergency management [1], [2], [6], [7]. Empirical studies show that higher predictive accuracy in these settings does not guarantee better human performance [3], [10].

Recent studies Work on responsible and explainable AI have emphasized the importance of stresses human-in-the-loop controls and governancestructures in AI development, particularly in safety-critical applications [2], [17], [18], [31]. However, most studies are either Most of this work, however, is conceptual or algorithm-centric and do not investigate human behavioral outcomes. There has been limited empirical research on the impact of specific AI system designs does not measure human behavior. Few studies have examined empirically how specific design features, such as uncertainty-aware escalation and guardrailson the execution of actions by humans , affect the actions humans take [1], [10].

By integrating these research streams, this research aims to place human action execution as a key criterion in the evaluation of AI-assisted decision systems and provide empirical contributions to

E. Information Sources for Disaster Decision Support

In disaster response, the input to the development of human-centered AI architectures that enhance their performance in high-stakes settings [1], [3]risk-modeling layer of ACHDF (Section III) comes from heterogeneous sensor networks, ad hoc communications, and community reporting, whose robustness affects the quality of the signal available for decision making [13]–[15]. Network-analytic methods such as community detection have been used to structure such large and noisy information environments (e.g., [32], [40]). These upstream methods are outside the scope of this paper. They are mentioned only because the uncertainty that ACHDF must represent originates in them.

III. System Overview and Architecture

This section is intended to introduce This section introduces the proposed human–AI decision system . In addition, it is intended to introduce the architectural elementsof the proposed human–AI decision system. The proposed human–AI decision system is intended to be and its architectural elements. The system is non-autonomous and human-centric. It is intended to have a balanced approach to human and algorithmic decision-making. The architectural elements of the proposed human–AI decision system are based on : the AI analyzes information and proposes options, and the human decides and acts. The architecture follows the **Action-Centered Human–AI Decision Framework (ACHDF)**. In the proposed framework, there is a significant emphasis on human action execution as the primary goal of the system, which, following the joint cognitive systems perspective [26], [27], treats the human operator and the AI as one decision system and takes appropriate human action as the system's goal.

A conceptual overview of the proposed action-centered architecture is illustrated in Fig. 1. As shown in the figure, ACHDF is organized into four primary design layers. The *environment and uncertainty context* layer captures incomplete, time-pressured, and asymmetric risk conditions that characterize high-stakes decision ecologies. Built on top of this, the *uncertainty-aware risk modeling* layer produces probabilistic risk estimates together with explicit confidence and uncertainty bounds. These estimates are then translated into constrained, policy-aligned options within the *guardrailed AI decision support* layer, which ensures that generated recommendations remain consistent with operational and safety constraints. Finally, the *human authority and accountability* layer preserves human control through oversight, override, and responsibility mechanisms. The two cross-cutting elements shown at the bottom of the figure—trust calibration , appropriate trust and cognitive ergonomics—continuously shape how information is presented so that operators can act reliably under uncertainty, are design considerations. They determined which cues are shown (e.g., explicit uncertainty indicators) and how much information is shown at once. The implemented system does not measure them or adapt to them at run time.

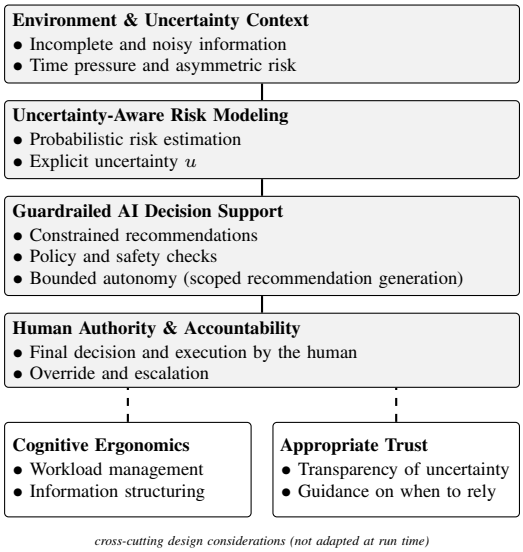


Fig. 1. Conceptual design layers of the Action-Centered Human–AI Decision Framework (ACHDF). The framework structures human–AI decision support into four layered components: environment and uncertainty context, uncertainty-aware risk modeling, guardrailed AI decision support, and preserved human authority and accountability. Appropriate trust [25] and cognitive ergonomics are cross-cutting design considerations that guided interface design. Trust and workload are also measured as evaluation outcomes. They are not adapted at run time (Table I).

The dynamic operationalization run-time behavior of these layers in real-time decision making is depicted in Fig. 2. This workflow view emphasizes how uncertainty-aware function allocation governs the interaction loop between AI and the human operator. When model confidence is high and risk is boundedWhen model uncertainty is low and the action is not critical, the system can provide presents a tightly scoped, action-oriented recommendations under bounded autonomy. As uncertaintyor potential impact increases, the escalation logic routes decisions to a human-in-the-loop review process, where recommendation. We call this mode *bounded autonomy*. The term refers only to the AI's freedom in *generating* a recommendation, which is bounded by guardrails. It does not mean that the AI executes actions or holds decision authority. When uncertainty, criticality, or a policy check exceeds its threshold, the system switches to *escalation*: instead of a single recommendation, it presents

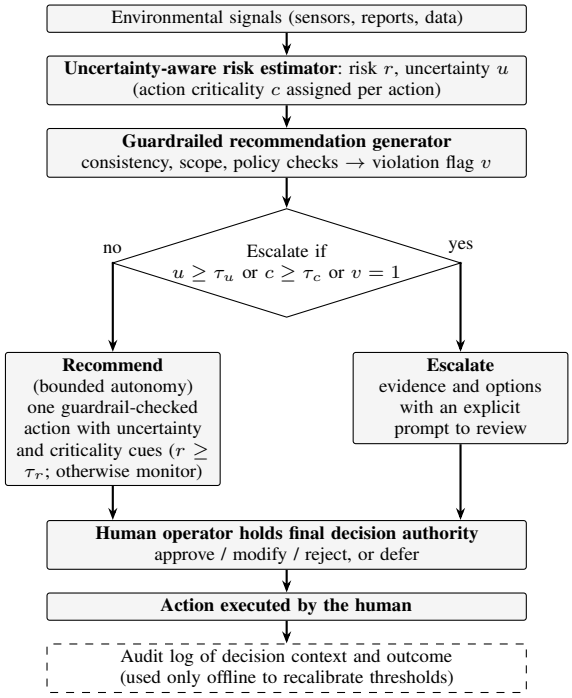


Fig. 2. Operational workflow of the proposed human–AI decision system. The escalation test follows Algorithm 1. Depending on the estimated uncertainty, the action criticality, and the guardrail policy checks, the system either presents a bounded recommendation or escalates the case for human review with evidence and options (when $r < \tau_r$ and no trigger fires, the system only monitors). In both modes, the human operator holds final decision and execution authority. Trust and workload are not adapted at run time. The audit log is used only offline.

the evidence and the available options and explicitly asks the operator to review them, and the operator can approve, modify, or reject the suggested action. The feedback paths shown in the figure highlight how action outcomes and operator responses contribute to ongoing trust calibration and workload management, forming a closed human–AI decision loop oriented toward reliable action execution rather than mere prediction display Thus, what varies between the two modes is the *level and form of AI support*. Final decision and execution authority stays with the human in both modes. Operator responses and outcomes are logged for audit and for offline recalibration of the thresholds.

A. Constructs and Their Roles

Table I summarizes the role of each ACHDF construct, i.e., whether it is a design input, a system state, a design consideration, or an evaluation outcome, together with its operational definition in the implemented system.

B. Design Goals and Human-in-the-Loop Principles

The design of the proposed system is informed by three key goals that are derived from the fundamental principles of ACHDF.

Goal 1 — — Enabling Human Action Execution. The proposed system must be able to system should support *human action execution* effectively, as opposed to simply only improving situational awareness and prediction accuracy. The key goal of the proposed system is therefore to enable effective human decision-making and action in operational settings.

Goal 2 — — Supporting Calibrated Appropriate Trust. The proposed system must system should support appropriate trust calibration [25] by making its uncertainty , model limitations, and potential failure modes explicit. By doing so, the proposed system enables users to develop a realistic understanding of when to trust the system's explicit, so that users can judge when to rely on its outputs and when to trust human rely on their own judgment.

Goal 3 - — Preserving Human Authority. The system shall ensure the preservation of should preserve human authority in high-risk decisionsby incorporating human-in-the-loop mechanisms and bounded autonomy. This means that, in safety-critical or ethically critical situations, the system will ensure that human operators have . Human operators always hold the final decision-making authority, avoiding over-reliance on automation or artificial intelligenceand execution authority.

To ensure that these goalsare achieved, the system will employ a human–AI collaboration paradigm, where the AI will be used to provide meet these goals, the AI provides decision scaffolding and action-oriented recommendations, and the human will have the final responsibility remains responsible for executing actions. The system will employ automation that assists in information synthesis and option generation, but will not be used for executing irreversible actions. This ensures the implementationof the concept of In the terms of Parasuraman *et al.* [28], the system automates information analysis and decision selection, but not action implementation. The level and form of support are adapted to the estimated uncertainty and the criticality of the action (uncertainty-governed function allocation, where the level of automation will be dynamically adapted according to the level of uncertainty and criticality of actionsadaptation of support).

C. Probabilistic Risk Estimation With Uncertainty

The heart of the system is a probabilistic The risk estimation module that provides calibrated risk assessments along provides risk assessments together with uncertainty representations. Unlike many models that produce point estimateresults, the model provides posterior risk distributions that include data-driven variability and epistemic uncertainty . Rather than a point estimate, the module outputs a posterior risk distribution, from which the risk level r and the uncertainty u are derived. This paper focuses on how the risk estimate and its uncertainty are used in the human–AI interaction, not on the risk model itself. The model's estimation procedure and calibration are therefore not evaluated here (Section VII-D).

Role 1 in ACHDF. These estimates of uncertainty are a crucial part of the decision logic. The **Role in ACHDF.** The risk level is interpreted in conjunction with the confidence intervals to determine together with its uncertainty to decide whether the system should make recommendations on its own or whether the process should be deferred to humans. The system's explicit representation of its own uncertainties helps to prevent the propagation of false precisionand supports the reasoning of the risk level in a way that mirrors the human decision-making processpresents a bounded recommendation or escalates (Algorithm 1). Explicitly representing uncertainty is intended to avoid conveying false precision.

D. Guardrailed AI Recommendation Module

To make risk estimates actionable and provide appropriate recommendations, a A generative AI module is used by the system. converts the risk estimate into a recommended action and a short justification.

Role 2 in ACHDF The system is constrained by a number of layers of structural guardrails that provide assurance with regard to reliability, consistency, and compliance with policies. These guardrails ensure that the recommendations generated by the

TABLE I
ACHDF CONSTRUCTS: ROLE, OPERATIONAL DEFINITION, AND IMPLEMENTATION

Construct	Role in ACHDF	Operational definition / implementation
Uncertainty u	Design input; determines system mode	Dispersion of the posterior risk estimate (Section III-C); compared with threshold τ_u.
Criticality c	Design input; determines system mode	Severity of the consequences if the recommended action is wrong or delayed (cost of error); compared with threshold τ_c.
Guardrails	Constraint on recommendation generation	Rule-based checks applied to every generated recommendation (Section III-D). A recommendation that fails a check is withheld and the case is escalated ($v = 1$ in Algorithm 1).
Bounded autonomy	System mode ("recommend")	The AI presents one scoped, guardrail-checked recommendation. The AI never executes actions; the human decides and acts.
Escalation	System mode ("escalate")	Triggered if $u \geq \tau_u$, $c \geq \tau_c$, or a policy check fails ($v = 1$). The system presents evidence and options and asks the operator to approve, modify, or reject.
Trust	Design consideration; secondary outcome	Shaped interface design (explicit uncertainty and criticality cues). Measured post-condition by questionnaire (Section IV-D). Not adapted at run time.
Workload	Design consideration; secondary outcome	Shaped interface design (single consolidated view, action-oriented phrasing). Measured post-condition by NASA–TLX [45]. Not adapted at run time.
Action execution	Primary outcome	Measured by AER (Section IV-D).

system align with the risk estimates that were made previously and that there is no contradiction with those estimates.There are also further constraints that restrict the scope of recommendations based on **Role in ACHDF.** Every generated recommendation passes through guardrails before it is shown: (i) a *consistency check* ensures that the recommended action does not contradict the risk estimate (e.g., no "no action needed" message when $r \geq \tau_r$); (ii) a *scope check* ensures that the recommendation is selected from the set of permitted actions for the scenario; and (iii) a *policy check* ensures compliance with operational and safety considerations. These recommendations also provide context cues that indicate the criticality and confidence levels associated with them, thereby avoiding information processing and facilitating a smooth transition to decisions and actions under pressurerules. A recommendation that fails any check is not shown; instead, the case is escalated ($v = 1$). The specific rule sets, the generative model configuration, and the scale and thresholds for criticality are not reported in this paper (Section VII-D). Each recommendation is displayed with its criticality and confidence cues. By bounding the generative results output within a structured decision process, the system harnesses the flexibility of generative AI while also limiting the risks of hallucination and overgeneralization, thereby ensuring that AI support is module aims to keep AI support assistive rather than prescriptive . and to limit hallucination and overgeneralization.

E. Human Override and Escalation Mechanisms

The escalation process, as described Escalation follows the policy in Algorithm 1, is linked to an uncertainty-aware policy that is used to dynamically allocate authority depending on the level of uncertainty and the criticality of the actions taken. The human override and escalation mechanism is an important part of the overall system architecture. The system automatically escalates decisions to human operators if the level of uncertainty is above certain thresholds, the potential impact of the recommendation is high, and if certain policy constraints are violated. In fact, escalation . The system escalates a case to the operator if *any* of three conditions holds: the uncertainty exceeds τ_u, the criticality of the action exceeds τ_c, or a guardrail policy check fails ($v = 1$). Escalation is not a failure modebut a decision to escalate to ensure accountability under uncertain conditions. The operators are free to either . It is a deliberate change in the form of support that keeps the operator accountable under uncertainty. In every mode, operators can accept, modify, or reject the decision

Algorithm 1 Uncertainty-aware escalation and human override policy

Require: risk estimate r, uncertainty u, action criticality c
Require: policy-violation flag $v \in \{0, 1\}$ from guardrail checks
Require: thresholds τ_r, τ_u, τ_c
if $u \geq \tau_u$ **or** $c \geq \tau_c$ **or** $v = 1$ **then**
  MODE $\leftarrow$ "escalate" ▷ route to human review
else if $r \geq \tau_r$ **then**
  MODE $\leftarrow$ "recommend" ▷ bounded autonomy
else
  MODE $\leftarrow$ "monitor"
end if
if MODE $=$ "escalate" **then**
  Present evidence and options to operator
  Operator selects: approve / modify / reject / defer
else if MODE $=$ "recommend" **then**
  Present guardrail-checked recommendation
  Operator decides and executes, overrides, or defers
end if
Log decision context and outcome for audit and calibration

recommended by the AIsystemAI's recommendation, and all of these operations are recorded for accountabilitypurposes. This minimization of processing of superfluous information is also beneficial for the management of cognitive workload for logged for accountability. Showing a single consolidated recommendation in routine cases is intended to reduce the amount of information operators must process in time-critical situations. This dynamic allocation of autonomy allows the system to offer effective assistance in regular situationsand, at the same time, allow human control adaptation of the level of support (a scoped recommendation in routine situations, a full review in uncertain or risky situations, thus striking a proper balance between performance and safety, as per the existing design principles for human–machine systems, which provides a platform for the ACHDF design principles proposed later in the papercritical ones) follows established principles of adaptive automation [28], [30] and does not transfer decision authority to the AI.

F. ACHDF Potential *Transferability Across Decision Ecologies*

Although the system architecture is instantiated and verified in the context of disaster response, the design principlesare not domain-specific in nature. On the contrary, the architecture presented here is more aligned with a general The architecture

TABLE II
HYPOTHESIZED MAPPING OF ACHDF DESIGN PRINCIPLES TO OTHER DECISION ECOLOGIES (NOT EMPIRICALLY TESTED)

Decision Domain	Primary Sources of Uncertainty	Asymmetry of Error Costs	Time Pressure Characteristics	Potential Role of Uncertainty-Aware Support Adaptation
Disaster Response (evaluated here)	Incomplete situational awareness, noisy sensor and report data, rapidly evolving conditions	Delayed action leads to loss of life or property; false alarms incur operational cost	Severe time pressure with dynamic escalation requirements	AI provides bounded, action-oriented recommendations under low uncertainty; high uncertainty triggers human-in-the-loop escalation to preserve accountability
Autonomous Driving Supervision	Perception uncertainty, rare edge cases, environment variability	False negatives may cause accidents; false positives reduce efficiency and trust	Sub-second to seconds-level response requirements	Low-uncertainty situations allow bounded recommendations; ambiguous or novel situations escalate to human supervision and override
Medical AI (Triage / Diagnosis)	Incomplete patient data, population-to-individual generalization limits	False negatives risk patient harm; false positives cause overtreatment	Moderate to high time pressure depending on clinical context	AI assists with risk stratification and option generation; high uncertainty or high criticality decisions defer to clinicians
Military Safety Automation	Adversarial uncertainty, incomplete intelligence, strategic deception	Incorrect action may cause catastrophic or irreversible consequences	High time pressure combined with strategic deliberation	AI supports sensemaking and constrained recommendations; final decision authority remains with human operators under uncertainty

was instantiated and evaluated only in disaster response. The design principles, however, address a structural class of *decision ecologies* that are characterized by high uncertainty, asymmetrical costs of errors, timing pressuresasymmetric costs of error, time pressure, and human accountability. The **Action-Centered Human–AI Decision Framework (ACHDF)** is intended to be more general in nature to cover a range of different safety-critical domains that have similar structural characteristics.

In order to further illustrate the general nature of the design principles presented in this work, TableII provides a decision ecology mapping that illustrates the transferability of the design principles to different application domains that are presented in this paper, such as the supervision of autonomous driving , medical AI in the context of medical Table II maps these characteristics to three other domains (autonomous driving supervision, medical triage or diagnostic supportsystems, or , and military safety automationsystems. Despite the varying nature of the uncertainty in each of the application domains considered in this paper, the role of) to show where uncertainty-aware function allocation is similar in nature. The contribution of this work is not about optimizing any particulardomain or application scenario in the context of human–AI decisions under uncertaintyadaptation of support *could* play a similar role. This mapping is a hypothesis for future work, not an empirical result. In particular, time scales, operator expertise, and regulatory constraints differ substantially across these domains (Section VII).

IV. EXPERIMENTAL DESIGN AND METHODS

In this section , we describe the design of the experiments that can be used to test the proposed human–AI decision system . In our case, we aimed to isolate the effect of AI support on human action execution, response behavior, and subjective experience. In general, our design of experiments follows the usual practice of designing experiments to evaluate human–machine systems, where experiments are defined with clear conditions for baseline and treatment cases This section describes the controlled user study that compared the ACHDF-based system with a conventional dashboard. The study is the only empirical evaluation reported in this paper. Details that could not be reported because the underlying materials or trial-level data were not available are listed in Section VII-D.

A. Study Design and Conditions *(Baseline vs. AI-Assisted)*

A We used a within-subjects design was employed in the experiment. In this design, the participants are presented with the same decision-making task in two different with two conditions: a *baseline* condition and an *AI-assisted* condition. In the baseline condition, the (ACHDF) condition. The order of the conditions was counterbalanced across participants to control for order and learning effects. Both conditions used the same underlying scenario data. Counterbalancing balances order effects across participants, but it does not remove carry-over: in the second condition, participants were presented with the conventional dashboard-based interface. In the AI-assisted condition, the participants were presented with the proposed system that provides context-aware recommendations through the human–AI decision architecture. To avoid learning effects and order effects in the experiment, the presentation order of the two conditions is counterbalanced. The significance of the results is determined by conducting a paired test. In both the baseline and AI-assisted conditions, the same data is used. Hence, the difference in the performance between the two conditions is solely due to the presence of AI assistance. may have remembered scenarios from the first. Because order effects could not be analyzed, the size of this carry-over is unknown (Section VII-D).

B. Participants and Sampling Strategy

The demographics of participants are presented in Table III**Baseline condition.** Participants used a conventional dashboard that displayed the hazard and risk information for each scenario. The dashboard provided no AI-generated action recommendation, uncertainty indicator, criticality cue, or escalation prompt. Participants had to integrate the information themselves and decide on an action.

AI-assisted condition. Participants used the ACHDF-based interface, which displayed the same underlying information together with the ACHDF features described in Section III: the risk estimate with an explicit uncertainty indicator, a criticality cue, and either a single guardrail-checked action recommendation ("recommend" mode) or, when escalation was triggered, the evidence and options with a prompt to review them ("escalate" mode). The decision logic is shown in Fig. 2 and Algorithm 1. Interface screenshots are not included (Section VII-D).

Because the AI-assisted condition differs from the baseline in several features at once (recommendation, action-oriented format, uncertainty cue, criticality cue, and

TABLE III
PARTICIPANT DEMOGRAPHICS

Characteristic	Value
Total participants	52
Age range	20s–50s
Gender	Mixed
Technical background	Low–high
Domain expertise required	No
Study design	Within-subjects

escalation), the comparison estimates the effect of the integrated ACHDF configuration. It does not isolate the effect of any single feature.

The total number of participants recruited for the study was 52 in number. For the purpose of this study, participants

B. Participants and Ethics

Fifty-two participants took part (Table III). Participants were recruited using the stratified sampling technique in order to create diversity in terms of stratified sampling to obtain diversity in age, gender, and technical know-how. This technique has been selected due to the heterogeneity observed in operators in real-world background, reflecting the heterogeneity of operators in real decision environments.

All participants gave their informed consent prior to the experiment. Although participants Participants were not required to have prior knowledge in the domain area, training on the task interface was provided to all participants to create uniformity in terms of familiarity with the proceduresdomain expertise, and the sample was not composed specifically of trained emergency-management professionals (Section VII).

The study was not submitted to an institutional review board or ethics committee for review. It involved only a simulated decision task and post-task questionnaires and interviews, with no physical or medical intervention. Participation was voluntary, and all participants gave informed consent before the experiment.

Sample size. No a priori power analysis was reported for this study. A post hoc sensitivity analysis shows that, with $n = 52$, a two-sided paired t-test at $\alpha = .05$ has 80% power to detect a within-subject effect of $d_z \approx 0.40$ or larger. Effects smaller than this may not have been detected reliably.

C. *Task Design and Procedure*

The participants were required to complete a set of time-sensitive

C. Decision Task and Procedure

Participants completed a series of time-limited decision tasks. These tasks involved understanding risk information and making a decision on whetheror not In each task, they interpreted risk information about an evolving hazard and decided whether, and how, to take a protective action. These decision tasks were aimed at capturing the essential aspects of a Participants could also defer the decision pending further confirmation; in the AI-assisted condition, escalation explicitly prompted such a review (deferral rates in both modes are reported in Table VII). The tasks were designed to reproduce the essential characteristics of high-stakes decision situation. These aspects include changes in decisions: changing information, uncertainty, and unbalanced asymmetric error costs. The experiment involved a training period, and after that, the participants completed the task execution of both the experimental conditions. The participants were

TABLE IV
OUTCOME MEASURES USED IN THE CONTROLLED STUDY

Measure	**Operationalization**
Action Execution Rate (AER)	Proportion of trials in which the participant executed the appropriate action within the specified time window (primary outcome).
Response Latency	Time from presentation of actionable information to action execution (s).
Cognitive Workload	NASA–TLX [45]; overall score and four subscales (0–100), post-condition.
Trust	Three items: perceived reliability, predictability, and appropriateness of recommendations (5-point Likert); overall score. Post-condition.
Usability	System Usability Scale (SUS) [46] (0–100), post-condition.

required Participants were instructed to focus on the efficacy of the taking the appropriate action rather than on the analysis of analyzing the information. The system was used to record the decision processes and interaction of the participants.

Before the experimental trials, all participants received training on the task interfaces so that they were equally familiar with the procedures. Participants then completed both conditions in counterbalanced order and completed the NASA–TLX, trust, and usability questionnaires after each condition. Short post-task interviews were held at the end of the session. The system logged all decisions, response times, and interactions.

D. *Measures and Metrics*

These measures and definitions are presented in

D. Measures

Table IV . To measure the effectiveness of the system from the perspective of human machine systems, several objective performancemeasures and subjective evaluation tools were used. summarizes the outcome measures.

Action Execution Rate. On the other hand, the Action Execution Rate (AER), which was defined as "*the proportion of tasks in which the participant performed the appropriate action within a specified time window,*" was also proposed as an indicator for the quality of support for not only good decision making but also good action execution, which was considered an important factor in the high-risk environmentThe AER is the proportion of trials in which the participant executed the *appropriate* action for the scenario within the specified time window. The AER therefore combines two aspects of performance: whether the action was correct and whether it was taken in time. Response latency, reported separately, describes the timing of all executed actions. The trial-level data needed to decompose the AER into a correct-action rate and an on-time rate, and to classify failures as wrong actions, late actions, or omissions, were not available for re-analysis. The procedure for defining the appropriate action and the length of the time window are also not documented in sufficient detail to be reported here. Both points are treated as limitations (Section VII-D).

Response Latency. Response latency was suggested as "the time it took since actionable information was displayed until a participant performed is the time from the display of actionable information until the participant executed an action. " Less response latency was considered to be more efficient decision support with less hesitation about uncertainty.

Cognitive Workload(NASA–TLX). Cognitive workload was measured using the NASA Task Load Index (NASA–TLX). The Workload was measured

with the NASA–TLX was administered to participants [45] after each condition, and it has multiple workload scales. Lower workload values represent lower mental demand and better cognitive efficiency. We report the overall score and four of the six subscales (mental demand, temporal demand, effort, and frustration); physical demand and performance were not reported. Lower scores indicate lower perceived workload. Whether the overall score is the raw (unweighted) or weighted TLX, and which subscales it includes, could not be verified (Section VII-D).

Trust and Usability. Trust in the system was determined through a post-task was measured after each condition with three Likert-type scale questionnaire evaluating perceived trustworthiness, predictability, and appropriateness items on the perceived *reliability*, *predictability*, and *appropriateness* of the system's suggestions. Usability was determined through a standardized usability questionnaire. These variables evaluate the subjective aspects that affect the adoption and use of human–AI systems. In addition to significance testing, effect sizes were calculated using Cohen's d to evaluate the practical significance of the differences found.

V. Results

This section presents the empirical resultsof the experimental evaluation. Following the conventions of the field of human–machine systems, the results are described with little interpretation.Statistical results and confidence intervals are provided as appropriate, and interpretation of the results is left to the following section. recommendations, rated on a 5-point scale. The overall trust score summarizes the three items. These three constructs are used consistently throughout the paper. Item-level data were not available, so internal consistency could not be estimated, and no published validation is available for these items, unlike established trust-in-automation scales such as that of Jian *et al.* [47]. The trust results should therefore be read as an indicative composite (Section VII-D).

A. *Action Execution and Response Time*

Participants using the AI-assisted condition had a higher Action Execution Rate compared to the baseline condition.In all scenarios, a larger percentage of participants were able to execute the correct action within a predetermined time window when assisted by the suggested system. Response latency was also found to be decreased in the AI-assisted condition . It was found that participants responded quicker on average after receiving actionable information. This shows that the suggested system helped participants transition quicker from risk interpretation to action execution. Summary statisticson Action Execution Rate and response latency are provided in Table **??**, and distributional differences are depicted in Fig. **??**. **Usability.** Usability was measured with the System Usability Scale (SUS) [46] after each condition. The SUS yields a score from 0 to 100, with higher scores indicating better perceived usability.

Qualitative observations. Post-task interviews were used to collect participants' comments on their experience. The comments were summarized into themes, but they were not systematically coded, and theme prevalence was not recorded. We therefore report them as informal observations that complement the quantitative results. We do not describe the study as a mixed-method evaluation.

A. *Task Performance Outcomes*

The performance in the execution of

A. Statistical Analysis

All comparisons are within-subjects (baseline vs. AI-assisted, paired by participant) and used paired significance tests. H1 (AER) is the objective task was enhanced with the help of AI, as shown in Fig. 3primary hypothesis; H2–H4 concern secondary outcomes. For each outcome, we report the condition means (and SDs, where available) and the difference between conditions. For the outcomes with reported SDs (NASA–TLX, trust, SUS), we computed Cohen's d_{av}, i.e., the mean difference divided by the average of the two condition SDs [48]. Because d_{av} does not take the within-subject correlation into account, it is not equivalent to d_z, the effect size on which the paired test is based. Exact test statistics, confidence intervals, assumption diagnostics, order-effect analyses, and multiplicity-corrected p-values could not be computed because the trial-level data were not available for re-analysis (Section VII-D). The comparisons between escalated and non-escalated trials (Section V-C) are descriptive only, because escalation is triggered by uncertainty or criticality and the two sets of trials are therefore not comparable.

V. Results

Table V reports the primary and secondary outcomes. Interpretation follows in Section VI.

It is observed that the rate of action execution was enhanced by a small amount, from 0.55

A. Action Execution and Response Latency

The mean AER was 0.62 in the baseline scenario to 0.74 condition and 0.81 in the AI-assisted scenario. This shows that the number of actions executed was higher in the AI-assisted scenario.

It is also observed that the response time was reduced. The average response time was reduced from 12.3 seconds condition, a difference of 0.19 (paired test, $p < .01$). The mean response latency was 12.4 s in the baseline scenario to 8.0 seconds condition and 8.1 s in the AI-assisted scenario. The time was reduced by more than 4 seconds. This shows that the participants responded and executed actions in a better manner with the help of the AI systemcondition, a difference of 4.3 s (paired test, $p < .01$). Latency distributions are not reported, because the trial-level data were not available (Section VII-D). Whether the difference in the AER arose mainly from fewer wrong actions or from fewer late actions could not be determined (Section VII-D).

B. *Cognitive* Workload *and* , Trust *Outcomes*, and Usability

A graphical representation of the results for workload, trust, and usability is shown in Fig. 3 . Cognitive workload, as assessed by NASA–TLX, was significantly lower in the AI-assisted condition than in the baseline condition. The total score on the summarizes the subjective measures, and Table VI gives the subscale scores. The overall NASA–TLX assessment dropped from around 60 score was 59.8 (SD 10.9) in the baseline condition to 46 and 46.2 (SD 9.6) in the AI-assisted condition, a reduction of 14 points . This is a clear indication of reduced workload in the AI-assisted condition. The NASA–TLX scores are summarized in Table **??**. Trust ratings were higher for the AI-assisted system. The average difference of 13.6 points (paired test, $p < .05$; $d_{av} = 1.32$). The overall trust score was higher in the AI-assisted condition , increasing from around 3.2 in the 3.4 (SD 0.6) in the baseline condition to 4.0 on a and 4.2 (SD 0.5) in the AI-assisted condition on the 5-point scale in the AI-assisted condition , indicating that

TABLE V
PRIMARY AND SECONDARY OUTCOMES: BASELINE VS. AI-ASSISTED (ACHDF), $n = 52$, WITHIN-SUBJECTS

Outcome	**Baseline** M (SD)	**AI-assisted** M (SD)	**Difference** (AI − baseline)	**Paired test**	**Effect size** d_{av}
Primary (H1)					
AER	0.62 (n/a)	0.81 (n/a)	+0.19	$p < .01$	n/a
Secondary (H2–H4)					
Response latency (s)	12.4 (n/a)	8.1 (n/a)	−4.3	$p < .01$	n/a
NASA–TLX overall (0–100)	59.8 (10.9)	46.2 (9.6)	−13.6	$p < .05$	1.32
Trust overall (1–5)	3.4 (0.6)	4.2 (0.5)	+0.8	$p < .05$	1.45
SUS (0–100)	68.5 (11.2)	81.7 (9.4)	+13.2	not reported	1.28

n/a: not available, because the trial-level data could not be re-analyzed. p-values are the significance thresholds of the original paired tests. Exact test statistics and 95% confidence intervals could not be computed (Section VII-D). d_{av} = mean difference / average of the two condition SDs, computed from the reported means and SDs. It ignores the within-subject correlation and is not equivalent to the paired d_z. A single Cohen's d of 0.86 given in the previous version could not be attributed to a specific outcome and has been removed.

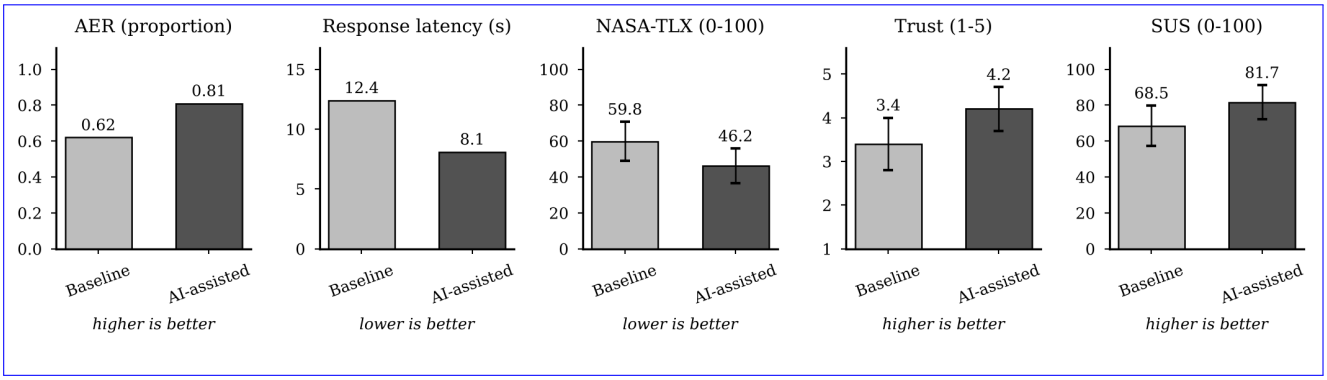


Fig. 3. Primary and secondary outcomes in the baseline and AI-assisted conditions (values from Table V). Each outcome is shown on its own scale. Error bars show ±1 SD where SDs are available (not available for AER and response latency).

TABLE VI
SUBSCALE SCORES, M (SD)

Subscale	**Baseline**	**AI-assisted**
NASA–TLX (0–100)		
Mental demand	63.4 (12.1)	48.7 (10.5)
Temporal demand	58.9 (11.4)	44.2 (9.8)
Effort	61.7 (13.0)	49.1 (11.2)
Frustration	55.3 (14.6)	41.8 (12.3)
Trust items (1–5)		
Perceived reliability	3.2 (0.7)	4.1 (0.6)
Perceived predictability	3.5 (0.8)	4.2 (0.6)
Perceived appropriateness	3.4 (0.6)	4.3 (0.5)

the system was perceived as more trustworthy and appropriate in its guidance. Usability was also higher (paired test, $p < .05$; $d_{av} = 1.45$). The SUS score was 68.5 (SD 11.2) in the baseline condition and 81.7 (SD 9.4) in the AI-assisted condition . The system usability scores increased from around 69 to 82, a jump of 13 points, indicating that the participants found the AI-assisted system interface easier and more effective to use. The trust and usability scores are summarized in Table **??**. ($d_{av} = 1.28$). No significance test was reported for the SUS.

C. *Effects of Uncertainty-Aware Escalation*

When the uncertainty thresholds were exceeded, human-in-the-loop escalation occurred. In such cases, the participants appropriately deferred their decisions until further confirmation or clarification was available. No increase in inappropriate use of the system recommendations in high-uncertainty cases was observed. The results comparing

C. Escalated and Non-Escalated Trials (Descriptive)

Table VII reports, descriptively, the rates of appropriate deferral and inappropriate reliance in escalated and non-escalated cases are presented in Table VII. These results show that uncertainty-aware escalation maintained system effectiveness while preventing over-reliance on the system.

TABLE VII
BEHAVIOR IN ESCALATED VS. NON-ESCALATED TRIALS, AI-ASSISTED CONDITION (DESCRIPTIVE)

Metric	**No escalation**	**Escalation triggered**
Appropriate deferral rate	0.21	0.47
Inappropriate reliance rate	0.18	0.09

trials of the AI-assisted condition. Appropriate deferral denotes deferring the decision for further confirmation when the situation warranted it. Inappropriate reliance denotes following the AI recommendation when this was not appropriate. These two sets of trials are not equivalent conditions. Escalation is triggered by high uncertainty or criticality, and it explicitly prompts human review. The differences in Table VII therefore cannot be attributed causally to escalation. The number of trials in each set was not available.

D. *Qualitative Observations*

The post-task interviews allowed for qualitative analysis of the participants' experiences. Participants often pointed out that the AI recommendations diminished the need for cross-referencing between different information sources. Some participants also emphasized the importance of having explicit uncertainty indicators to guide their confidence in decisions. Some qualitative findings are presented in

TABLE VIII
INFORMAL OBSERVATIONS FROM POST-TASK INTERVIEWS (NOT SYSTEMATICALLY CODED)

Theme	Observation (paraphrased)
Reduced context switching	Recommendations reduced the need to cross-reference multiple sources.
Usefulness of uncertainty cues	Participants used uncertainty cues to decide when to defer or seek confirmation.
Accountability and control	The ability to override improved perceived safety and responsibility.

D. Qualitative Observations

Table VIII . These findings supplement the quantitative findings by providing a glimpse into the participants' experiences with the system during decision-making tasks. To mitigate the potential for a Type I error due to multiple comparisons, all primary hypotheseswere defined a priori, and consistent performance patterns were verified across different objective and subjective measures.

summarizes recurring comments from the post-task interviews. These are informal observations. They were not systematically coded, and their prevalence was not recorded (Section IV-D).

VI. Discussion

This section interprets the results from a human–machine systems perspective. In line with the JCS view [26], [27], we interpret the results at the level of the joint human–AI system. Because the AI-assisted condition differed from the baseline in several features at once, the interpretations below about *which* features produced the effects are hypotheses, not demonstrated causal pathways.

A. Action Execution in the Joint Human–AI System

In the AI-assisted condition, the AER was higher and response latency shorter than with the dashboard. This is consistent with H1 and H2. Evaluating the system by action execution, rather than by prediction accuracy, follows directly from the JCS perspective. In AI decision-support research, however, evaluation is still dominated by accuracy and information quality [3], [10], [11]. Because the baseline offered no AI support at all, the comparison shows the benefit of the integrated ACHDF support over a dashboard. It does not show how much of this benefit is due to the way support was structured rather than to the availability of AI support as such; separating the two requires an additional condition with unstructured AI recommendations.

Several features of the AI-assisted condition could explain the improvement: the presence of a recommendation, its action-oriented phrasing, the explicit uncertainty and criticality cues, and escalation. Our design does not separate these features. In particular, we cannot conclude that presenting recommendations in an action format, rather than a risk format, caused the improvement. A factorial design that varies the format while holding the other features constant is needed to test this (Section VII). Because the AER could not be decomposed, it also remains open whether the improvement came mainly from fewer wrong actions or from fewer late actions (Section VII-D).

B. *Mechanism and Multi-layer Analysis Under ACHDF*

In order to provide a structured understanding of how the achieved results were obtained at the behavioral level, we also analyzed the results through the prism of the Action-Centered Human–AI Decision Framework (ACHDF). This analysis does not include any new outcome measures but structures the existing results to reveal multi-layered patterns relevant to human action in the context of uncertainty. From the mechanism perspective, the increase in Action Execution Rate and the reduction in response latency (Table **??**, Fig. **??**)may be considered a behavioral outcome of uncertainty-aware function allocation, where the representation of uncertainty directly influences the allocation of decisions to the AI or human operators in terms of timing and correctness of action execution. At the level of error structure,

B. Framework-Based Interpretation of the Results

The following interpretation organizes the observed results along the ACHDF layers. It does not test the proposed mechanisms; it is a framework-based reading of the existing outcomes. *Action level:* the higher AER and lower latency (Table V) are consistent with the idea that uncertainty-aware escalation was found to influence participants' responses in high-uncertainty scenarios. As depicted in Table VII, escalation was linked to increased appropriate deferral and reduced inappropriate reliance adaptation of support helps operators move from risk interpretation to action. *Error-structure level:* in escalated trials, appropriate deferral was more frequent and inappropriate reliance less frequent than in non-escalated trials (Table VII). Because escalation is triggered by uncertainty and criticality and itself prompts human review, this pattern is descriptive. It is compatible with, but does not demonstrate, a protective effect of escalation against over-reliance. These results indicate that uncertainty-aware escalation selectively mitigated failure modes related to hesitation and over-reliance, rather than indiscriminately increasing reliance on automation. Lastly, qualitative results *Process level:* interview themes on reduced context switching (Table VIII) and reductions in workload lower NASA–TLX scores (Table **??**) indicate changes in decision processes in VI) suggest that the AI-assisted condition . Participants' self-reported context switching and decision transitions align with reductions in response latency and workload. These findings on all three levels align with the premise of ACHDF: that human action in uncertain situations can be explained by the interaction of uncertainty-aware risk representation, bounded autonomy, calibrated trust, and cognitive workload management. Moreover, statistical tests for comparing the baseline and AI-assisted conditions revealed that the improvement in the rate of action execution and the reduction of latency while responding were statistically significant at a p value of less than 0.01.Differences observed for cognitive workload and trust levels, as measured by the NASA–TLX scale, were statistically significant at a p value of less than 0.05changed the decision process.

Workload as a possible explanation. Lower workload is a plausible contributor to better action execution. The NASA–TLX, however, is a retrospective, subjective, condition-level measure, and we did not test whether the reduction in workload explains the improvement in AER (e.g., with a mediation analysis). This link should therefore be regarded as a possible explanatory mechanism, not a demonstrated pathway.

VII. Discussion

This section interprets the empirical findings from a human–machine systems perspective and situates them within the broader literature on human–AI decision making. Rather than focusing on application-specific results, the discussion emphasizes generalizable

insights for the design and evaluation of human–AI decision systems operating under uncertainty [1]–[3], [10]. In particular, this work contributes to the core concerns of *Human–Machine Systems* by explicitly linking uncertainty-aware function allocation to observable human action execution. By doing so, it extends classic HMS constructs—such as trust calibration , workload, and authority allocation—from internal cognitive or perceptual states to measurable behavioral outcomes.

A. Trust and Automation Bias

Perceived trust was higher in the AI-assisted condition. Higher trust is not the same as *calibrated* trust. Calibration requires that reliance matches the actual reliability of the advice [25]. Showing calibration would require relating trust or reliance to the correctness of the AI's recommendations, trial by trial. The greater deferral in escalated trials is also not evidence of calibration, because escalation itself prompts human review. We therefore make no claim that ACHDF calibrates trust.

B. *Implications for Human–AI Decision Systems*

The results show that human performance gain is not simply a function of the availability of AI support, but also depends on how that support is organized to guide human action execution. By directly assessing action execution as an outcome measure, this research extends previous research that primarily tested decision support systems in terms of the accuracy of predictions or the quality of information [3], [10], [11]. From a human–machine system perspective, action execution is the end result of human information processing that combines elements of perception, understanding, trust, and workload into system performance [1], [12]. The increase in Action Execution Rate and the decrease in response latency indicate that human–AI systems are capable of significantly influencing human behavior when recommendationsare presented in action format instead of risk format. This supports the move towards an outcome-oriented human–machine system research area where the effectiveness of AI support is determined in terms of human action instead of computer performance [3], [10]. This research contributes to the Human–Machine Systems area by empirically connecting function allocation with uncertainty to human action execution. The results demonstrate how design decisions in the architecture (bounded autonomy, uncertainty representation, and escalation strategies) map to behavioral phenomena, thus closing the theoretical-practice divide in human factors theory and human–AI system design [1], [19], [31]. In this respect, the Action-Centered Human–AI Decision Framework (ACHDF) may also be used as an analytical tool for identifying failure modes in existing human–AI systems, aside from the system examined in this research [3], [10]. Similarly, automation bias can be directly assessed only on trials in which the AI's advice is wrong, by measuring how often participants accept it [23]. The study did not analyze behavior on trials with erroneous AI recommendations, so it cannot show whether ACHDF reduces automation bias. This is an important topic for future work.

B. *Trust Calibration and Automation Bias*

The main contribution of this research is in the way it approaches the issue of trust calibration. The findings show that the more trust participants had in the AI-assisted system, the more this did not translate to automation bias. Instead, participants displayed appropriate trust in the system and delayed action in situations of high uncertainty

B. Ecological Validity of Escalation

In the study, participants could defer a decision when the system escalated. In real time-critical emergencies, deferral may not be feasible or may itself be costly: an operator may have to act on uncertain information before an evacuation window closes. The lower inappropriate reliance observed in escalated trials may therefore not carry over to settings where deferral is constrained, and appropriate use of human-in-the-loop escalation mechanisms [17], [21], [22]. The findings demonstrate the effect of uncertainty-aware design and the use of design constraints in avoiding typical automation failure modes. In this way, the system did not provide false confidence to the participants. These findings validate human factors theories emphasizing the importance of trust calibration in effective forced action under uncertainty might change both reliance and workload. Future studies should assign explicit costs to deferral and include forced-choice conditions.

TABLE IX
OBSERVED FINDINGS AND CANDIDATE DESIGN IMPLICATIONS (TO BE TESTED COMPONENT-WISE)

Observed finding	Candidate design implication
Higher AER, lower latency	Action-oriented recommendations
Lower NASA–TLX	Consolidated information structuring
Higher perceived trust	Explicit uncertainty and criticality cues

C. Design Implications

Within the scope of the evaluated task, the results suggest that human–AI cooperation and provide evidence of this in design [1], [12]. These results provide empirical support for long-held theoretical assertions in human–machine systems research that automation bias can be reduced by incorporating uncertainty representation and decision support systems may benefit from (1) representing uncertainty explicitly, (2) providing action guidance rather than only passive information display, and (3) adapting the level and form of support to uncertainty and criticality through human-in-the-loop escalation. Contrary to the potential for increasing over-reliance, the proposed design facilitated calibrated trust by aligning perceived system confidence with actual uncertainty [10], [17], [18].

D. *Design Implications for High-Risk Domains*

Although the empirical study was carried out in the context of disaster response, the design principles and results can be generalized to other high-risk domains, while leaving final authority with the human [1], [19], [28], [31]. Table IX lists these candidate implications. Because the components were not manipulated separately, each implication should be tested individually. Whether they transfer to other safety-critical domains, such as healthcare, transportation, and industrial safety , where the cost of inappropriate or delayed response is high, and complete automation is not always feasible or desirable [2], [4], [6], [7]. Similar action-centered constraints and explainability considerations also arise in cyber-physical production and Industry 4.0 security decision support [49]. The findings of the study indicate that human–AI decision-making systems should be designed in such a way that they (1) model uncertainty explicitly, (2) provide action guidance rather than passive information display, and (3) allocate decision authority dynamically through human-in-the-loop escalation [1], [19], [31]. Based on the Human–Machine Systems model, the results of the current study imply that effective decision systems with AI assistance need to be based on the concepts of bounded autonomy and decision authority allocation, as well as uncertainty-aware interaction. These concepts are not limited to the disaster response domain and are generally applicable to a wide variety of safety-critical human–machine systems with AI assistance [2], [3]. The observed effects of the described phenomenon are not domain-specific but rather general mechanisms of human–machine interaction, which are relevant to any safety-critical decision system [1], [12]. In general, the current study contributes to the broader area of human–machine systems by showing the benefits of the application of human-centered AI decision systems with awareness of uncertainty for the execution of actions without compromising safety and accountability [3], [10]. The design implications that are critical based on the findings are presented in

Table IX. In providing insight into the manner in which architectural design translates into behavioral outcomes, the results contribute to the domain of human machine systems with respect to trust calibration and cognitive ergonomics. [2], [49], has not been tested here (Table II).

VII. Limitations and Future Work

The research has several limitations, which also point to potential directions for future research in similar studies. Although the research provides valuable data in terms of the effectiveness of the human–AI system in decision-making, it must be considered in relation to the limitations of this research.

A. *Generalization*

The study has conducted an empirical evaluation using a set of decision scenarios in relation to a disaster response scenario. Although this scenario is representative of

A. Bundled Intervention

The AI-assisted condition differed from the baseline in several features at once. The study therefore estimates the effect of the integrated ACHDF configuration and cannot attribute effects to individual components (uncertainty display, action format, escalation, guardrails). Future work should use factorial or ablation designs.

B. Trust Calibration and Automation Bias

Trust was measured as a self-reported attitude at the condition level, and behavior on trials with erroneous AI recommendations was not analyzed. Hence, neither calibration nor automation bias could be assessed directly. Future studies should include trials with incorrect advice and measure reliance trial by trial.

C. Domain and Population Generalizability

The evaluation was conducted only with disaster-response scenarios in a high-stakes decision-making scenario in which there is considerable uncertainty, future research can be conducted to evaluate the generalization of the proposed design principles across diverse domains and professionals in which there is considerable expertisesimulated setting. Participants were not required to have domain expertise, and the sample was not composed specifically of emergency-management professionals. Trained emergency operators may behave differently: they may rely less on recommendations, use different heuristics, or experience different workload. Generalization to other domains (e.g., medical triage, autonomous driving supervision, traffic management) and to professional operators remains future work. The ecological limits of deferral (Section VI) also belong here.

D. Reporting and Analysis Limitations

Several analyses and details requested for a complete account of the study could not be provided, because the trial-level data and some study materials were not available for re-analysis. The affected results should be interpreted with corresponding caution:

- *Inferential statistics.* Only the significance thresholds of the original paired tests are available. Exact test statistics, 95% confidence intervals, assumption diagnostics, the paired effect size d_z, order-effect analyses, and multiplicity-corrected p-values could not be computed. SDs and distributions for the AER and response latency are not available, and the paired test used for each outcome is not documented. Because both conditions used the same scenarios, carry-over from the first to the second condition cannot be ruled out, and its size could not be estimated without order-effect analyses. No a priori power analysis was reported; the sensitivity analysis (Section IV-B) indicates adequate power only for effects of about $d_z \geq 0.40$.
- *AER construction.* The procedure for defining the appropriate action and the length of the time window are not documented in detail, and the AER could not be decomposed into action correctness and timeliness. Consequently, it cannot be determined whether execution failures were due to wrong actions or to delays.
- *Task and materials.* The number and composition of scenarios (by uncertainty and criticality), the full set of response options, the per-trial time limit, the matching of scenario difficulty, the training content and any pass criterion, and interface screenshots are not reported. This limits reproducibility.
- *System implementation.* The risk model's estimation and calibration procedure, the criticality scale and thresholds, and the specific guardrail rules are not reported. The evaluation therefore concerns the interaction design as experienced by participants, not the validity of these components.
- *Measures.* The trust items have no published validation and their internal consistency could not be estimated. Only four of the six NASA–TLX subscales are reported, and the overall score cannot be fully reconciled with them: it equals the mean of the four subscales in the baseline condition (59.8) but not in the AI-assisted condition (46.2 vs. 46.0), so the scoring procedure could not be verified. The interview data were not systematically coded. The number of escalated trials is not available.

Future studies should preregister their analyses and share trial-level data and materials so that these analyses can be reported in full.

E. *Long-Term Trust Effects*

The research has focused on the study captured only short-term effects of human–AI systems. Although it is recognized that the study has focused on the short term, it is anticipated that the long-term effects of trust will be considerable in relation to operators' within a single session. Trust and reliance may change with longer exposure to the system and its behavior in relation to uncertainty escalation .

F. *Extension of the Research Beyond Disaster Response*

The research has focused on the domain of disaster response and has recognized this domain as effective in relation to testing systems in a high-stakes scenario. Although the domain is recognized as effective in relation to testing systems in a high-stakes scenario, future research can be conducted to extend the system beyond this domain to other high-stakes domains such as medical systems and traffic management systemsto its escalation behavior, which calls for longitudinal field studies.

VIII. Conclusion

This work evaluated paper presented ACHDF, an action-centered human–AI decision system for disaster-risk contexts and showed that an ACHDF-based, uncertainty-aware architecture improves action execution (higher AER and lower response latency) while reducing cognitive workload(NASA–TLX) and supporting calibrated trustwithout increasing automation bias. By empirically linking framework that instantiates the joint cognitive systems perspective for disaster-response decision support. ACHDF combines uncertainty-aware function allocation, guardrailed recommendations, risk estimation, guardrailed recommendations, and uncertainty- and human-in-the-loop escalation to observable action execution, the paper advances core concerns of human–machine systems and provides transferable criticality-triggered escalation, while the human keeps final decision authority. In a within-subjects study with 52 participants, the integrated ACHDF configuration, compared with a conventional dashboard, was associated with a higher action execution rate, shorter response latency, lower self-reported workload, and higher perceived trust; usability scores were also higher, but without a reported significance test. Only significance thresholds were available for the inferential results. The study did not isolate individual components, assess trust calibration or automation bias directly, or test other domains or professional operators. These questions define the next steps toward validated design principles for other safety-critical decision ecologieshuman–AI decision support under uncertainty.

Acknowledgment

The author thanks the municipal partners who supported the pilot deployment and the study participants for their time and insights. The author thanks the colleagues in the areas of human machine , and colleagues in human–machine systems and human-centered AI research for the valuable discussionsthat contributed to the development of this workvaluable discussions.

References

[1] T. A. Bach, J. K. Kristiansen, A. Babic, and A. Jacovi, "Unpacking Human-AI Interaction in Safety-Critical Industries: A Systematic Literature Review," *IEEE Access*, vol. 12, pp. 106 385–106 414, 2024. [Online]. Available: https://doi.org/10.1109/ACCESS.2024.3437190

[2] J. Perez-Cerrolaza, J. Abella, M. Borg, C. Donzella, J. Cerquides, F. J. Cazorla, C. Englund, M. Tauber, G. Nikolakopoulos, and J. L. Flores, "Artificial Intelligence for Safety-Critical Systems in Industrial and Transportation Domains: A Survey," *ACM Computing Surveys*, vol. 56, no. 7, 2024. [Online]. Available: https://doi.org/10.1145/3626314

[3] C. Yang, Y. Zhu, and Y. Chen, "A Review of Human–Machine Cooperation in the Robotics Domain," *IEEE Trans. Human-Machine Systems*, vol. 52, no. 1, pp. 12–25, 2022. [Online]. Available: https://doi.org/10.1109/THMS.2021.3131684

[4] J. Birkmann, O. D. Cardona, M. L. Carreño, and et al., "Framing Vulnerability, Risk and Societal Responses: The MOVE Framework," *Natural Hazards*, vol. 67, pp. 193–211, 2013. [Online]. Available: https://doi.org/10.1007/s11069-013-0558-5

[5] V. Sandoval, M. Voss, V. Flörchinger, and et al., "Integrated Disaster Risk Management (IDRM): Elements to Advance Its Study and Assessment," *International Journal of Disaster Risk Science*, vol. 14, pp. 343–356, 2023. [Online]. Available: https://doi.org/10.1007/s13753-023-00490-1

[6] M. Rokhideh, C. Fearnley, and M. Budimir, "Multi-Hazard Early Warning Systems in the Sendai Framework for Disaster Risk Reduction: Achievements, Gaps, and Future Directions," *International Journal of Disaster Risk Science*, vol. 16, pp. 103–116, 2025. [Online]. Available: https://doi.org/10.1007/s13753-025-00622-9

[7] S. Rao, F. Doherty, A. Traver, and et al., "Extreme Weather Disruptions and Emergency Preparedness Among Older Adults in Ohio: An Eight-County Assessment," *International Journal of Disaster Risk Science*, vol. 15, pp. 213–225, 2024. [Online]. Available: https://doi.org/10.1007/s13753-024-00548-8

[8] P. Charan, M. M. Siddiqui, V. Yadav, C. Pathak, Y. Narayan, and Z. H. Khan, "AI-Enhanced Early Warning Systems for Natural Disaster Detection and Mitigation using Wireless Sensor Networks," in *Proc. Int. Conf. Computational and Characterization Techniques in Engineering & Sciences (IC3TES)*, 2024, pp. 1–6. [Online]. Available: https://doi.org/10.1109/IC3TES62412.2024.10877562

[9] G. Lim, "Technical Report: A Generative AI-Driven Reliability Layer for Action-Oriented Disaster Resilience," 2026. [Online]. Available: https://arxiv.org/abs/2601.18308

[10] V. Lai, C. Chen, A. Smith-Renner, Q. V. Liao, and C. Tan, "Towards a Science of Human-AI Decision Making: An Overview of Design Space in Empirical Human-Subject Studies," in *Proc. ACM Conf. Fairness, Accountability, and Transparency*, 2023, pp. 1369–1385. [Online]. Available: https://doi.org/10.1145/3593013.3594087

[11] M. Muller and J. Weisz, "Extending a Human-AI Collaboration Framework with Dynamism and Sociality," in *Proc. Symp. Human-Computer Interaction for Work*, 2022. [Online]. Available: https://doi.org/10.1145/3533406.3533407

[12] A. Hodgson, C. E. Siemieniuch, and E.-M. Hubbard, "Culture and the Safety of Complex Automated Sociotechnical Systems," *IEEE Trans. Human-Machine Systems*, vol. 43, no. 6, pp. 608–619, 2013. [Online]. Available: https://doi.org/10.1109/THMS.2013.2285048

[13] S. H. Stephens and A. Altamirano, "Supporting Community Resilience to Environmental Hazards through User-Centered Design," in *Proc. ACM Int. Conf. Design of Communication*, 2022, pp. 125–129. [Online]. Available: https://doi.org/10.1145/3513130.3558989

[14] G. Longo, A. Cantelli-Forti, E. Russo, F. Lupia, M. Strohmeier, and A. Pugliese, "Collective Victim Counting in Post-Disaster Response: A Distributed, Power-Efficient Algorithm via BLE Spontaneous Networks," *Pervasive and Mobile Computing*, vol. 106, 2025. [Online]. Available: https://doi.org/10.1016/j.pmcj.2024.101997

[15] H. Yang, F. Ye, Y. Yuan, S. Lu, and W. Arbaugh, "Toward Resilient Security in Wireless Sensor Networks," in *Proc. ACM Int. Symp. Mobile Ad Hoc Networking and Computing*, 2005, pp. 34–45. [Online]. Available: https://doi.org/10.1145/1062689.1062696

[16] R. McAllister, Y. Gal, A. Kendall, M. Van Der Wilk, A. Shah, R. Cipolla, and A. Weller, "Concrete Problems for Autonomous Vehicle Safety: Advantages of Bayesian Deep Learning," in *Proc. Int. Joint Conf. Artificial Intelligence*, 2017, pp. 4745–4753.

[17] L. Weidinger, J. Uesato, M. Rauh, C. Griffin, P.-S. Huang, J. Mellor, A. Glaese, M. Cheng, B. Balle, A. Kasirzadeh, C. Biles, S. Brown, Z. Kenton, W. Hawkins, T. Stepleton, A. Birhane, L. A. Hendricks, L. Rimell, W. Isaac, J. Haas, S. Legassick, G. Irving, and I. Gabriel, "Taxonomy of Risks Posed by Language Models," in *Proc. ACM Conf. Fairness, Accountability, and Transparency*, 2022, pp. 214–229. [Online]. Available: https://doi.org/10.1145/3531146.3533088

[18] T. Miller, "But Why? Understanding Explainable Artificial Intelligence," *XRDS*, vol. 25, no. 3, pp. 20–25, 2019. [Online]. Available: https://doi.org/10.1145/3313107

[19] X.-M. Zhang and H.-N. Wu, "Multiobjective Control Design for Human–Machine Systems With Safety Performance Constraints," *IEEE Trans. Human-Machine Systems*, vol. 52, no. 4, pp. 636–647, 2022. [Online]. Available: https://doi.org/10.1109/THMS.2021.3116123

[20] G. Mai, W. Huang, J. Sun, S. Song, D. Mishra, N. Liu, S. Gao, T. Liu, G. Cong, Y. Hu, C. Cundy, Z. Li, R. Zhu, and N. Lao, "On the Opportunities and Challenges of Foundation Models for GeoAI (Vision Paper)," *ACM Trans. Spatial Algorithms and Systems*, vol. 10, no. 2, 2024. [Online]. Available: https://doi.org/10.1145/3653070

[21] I. D. Raji and J. Buolamwini, "Actionable Auditing: Investigating the Impact of Publicly Naming Biased Performance Results of Commercial AI Products," in *Proc. AAAI/ACM Conf. AI, Ethics, and Society*, 2019, pp. 429–435. [Online]. Available: https://doi.org/10.1145/3306618.3314244

[22] ——, "Actionable Auditing Revisited: Investigating the Impact of Publicly Naming Biased Performance Results of Commercial AI Products," *Communications of the ACM*, vol. 66, no. 1, pp. 101–108, 2022. [Online]. Available: https://doi.org/10.1145/3571151

[23] R. Parasuraman and D. H. Manzey, "Complacency and Bias in Human Use of Automation: An Attentional Integration," *Human Factors*, vol. 52, no. 3, pp. 381–410, 2010. [Online]. Available: https://doi.org/10.1177/0018720810376055

[24] L. Duan, W. N. Street, Y. Liu, and H. Lu, "Community Detection in Graphs through Correlation," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2014, pp. 1376–1385. [Online]. Available: https://doi.org/10.1145/2623330.2623629

[25] J. D. Lee and K. A. See, "Trust in Automation: Designing for Appropriate Reliance," *Human Factors*, vol. 46, no. 1, pp. 50–80, 2004. [Online]. Available: https://doi.org/10.1518/hfes.46.1.50_30392

[26] E. Hollnagel and D. D. Woods, *Joint Cognitive Systems: Foundations of Cognitive Systems Engineering*. Boca Raton, FL, USA: CRC Press, 2005.

[27] D. D. Woods and E. Hollnagel, *Joint Cognitive Systems: Patterns in Cognitive Systems Engineering*. Boca Raton, FL, USA: CRC Press, 2006.

[28] R. Parasuraman, T. B. Sheridan, and C. D. Wickens, "A Model for Types and Levels of Human Interaction With Automation," *IEEE Trans. Syst., Man, Cybern. A, Syst. Humans*, vol. 30, no. 3, pp. 286–297, 2000. [Online]. Available: https://doi.org/10.1109/3468.844354

[29] D. B. Kaber and M. R. Endsley, "The Effects of Level of Automation and Adaptive Automation on Human Performance, Situation Awareness and Workload in a Dynamic Control Task," *Theoretical Issues in Ergonomics Science*, vol. 5, no. 2, pp. 113–153, 2004. [Online]. Available: https://doi.org/10.1080/1463922021000054335

[30] M. W. Scerbo, "Adaptive Automation," in *Neuroergonomics: The Brain at Work*, R. Parasuraman and M. Rizzo, Eds. New York, NY, USA: Oxford University Press, 2007, pp. 239–252.

[31] Z. Assaad and C. Boshuijzen-van Burken, "Ethics and Safety of Human-Machine Teaming," in *Proc. Int. Symp. Trustworthy Autonomous Systems*, 2023. [Online]. Available: https://doi.org/10.1145/3597512.3600205

[32] K. Wakita and T. Tsurumi, "Finding Community Structure in Mega-Scale Social Networks: Extended Abstract," in *Proc. Int. World Wide Web Conf.*, 2007, pp. 1275–1276. [Online]. Available: https://doi.org/10.1145/1242572.1242805

[33] L. Prokhorenkova and A. Tikhonov, "Community Detection through Likelihood Optimization: In Search of a Sound Model," in *Proc. World Wide Web Conf.*, 2019, pp. 1498–1508. [Online]. Available: https://doi.org/10.1145/3308558.3313429

[34] Y. Xian, P. Li, H. Peng, Z. Yu, Y. Xiang, and P. S. Yu, "Community Detection in Large-Scale Complex Networks via Structural Entropy Game," in *Proc. ACM Web Conf.*, 2025, pp. 3930–3941. [Online]. Available: https://doi.org/10.1145/3696410.3714837

[35] G. M. Slota, J. W. Berry, S. D. Hammond, S. L. Olivier, C. A. Phillips, and S. Rajamanickam, "Scalable Generation of Graphs for Benchmarking HPC Community-Detection Algorithms," in *Proc. Int. Conf. High Performance Computing, Networking, Storage and Analysis*, 2019. [Online]. Available: https://doi.org/10.1145/3295500.3356206

[36] P. G. Sun, "Analysis of Resolution Limit in Community Detection," in *Proc. Int. Conf. Fuzzy Systems and Knowledge Discovery (FSKD)*, 2014, pp. 761–765. [Online]. Available: https://doi.org/10.1109/FSKD.2014.6980932

[37] A. Rezaei, S. M. Far, and M. Soleymani, "Near Linear-Time Community Detection in Networks with Hardly Detectable Community Structure," in *Proc. IEEE/ACM Int. Conf. Advances in Social Networks Analysis and Mining (ASONAM)*, 2015, pp. 65–72. [Online]. Available: https://doi.org/10.1145/2808797.2808903

[38] S. Akhter, S. Asif, M. H. Sazzad, and M. M. Hasan, "Performance Analysis of Parallel Overlapping Community Detection Algorithms in Large-Scale Social Networks," in *Proc. Int. Conf. Computing Advancements*, 2022, pp. 34–39. [Online]. Available: https://doi.org/10.1145/3542954.3543044

[39] P. G. Sun, L. Gao, and S. Han, "Identification of Overlapping and Non-Overlapping Community Structure by Fuzzy Clustering in Complex Networks," *Information Sciences*, vol. 181, no. 6, pp. 1060–1071, 2011. [Online]. Available: https://doi.org/10.1016/j.ins.2010.11.022

[40] T. C. Havens, J. C. Bezdek, C. Leckie, K. Ramamohanarao, and M. Palaniswami, "A Soft Modularity Function for Detecting Fuzzy Communities in Social Networks," *IEEE Trans. Fuzzy Systems*, vol. 21, no. 6, pp. 1170–1175, 2013. [Online]. Available: https://doi.org/10.1109/TFUZZ.2013.2245135

[41] C. Vehlow, T. Reinhardt, and D. Weiskopf, "Visualizing Fuzzy Overlapping Communities in Networks," *IEEE Trans. Visualization and Computer Graphics*, vol. 19, no. 12, pp. 2486–2495, 2013. [Online]. Available: https://doi.org/10.1109/TVCG.2013.232

[42] Y. Zhang and D.-Y. Yeung, "Overlapping Community Detection via Bounded Nonnegative Matrix Tri-Factorization," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2012, pp. 606–614. [Online]. Available: https://doi.org/10.1145/2339530.2339629

[43] K. Cao, Y. Liu, and H. Zhang, "Fuzzy Modularity Based Louvain Algorithm for Overlapping Community Detection," in *Proc. Int. Conf. Intelligent Information Processing*, 2020, pp. 18–23. [Online]. Available: https://doi.org/10.1145/3378065.3378069

[44] C. Li, H. Chen, T. Li, and X. Yang, "A Stable Community Detection Approach for Complex Network Based on Density Peak Clustering and Label Propagation," *Applied Intelligence*, vol. 52, no. 2, pp. 1188–1208, 2022. [Online]. Available: https://doi.org/10.1007/s10489-021-02287-5

[45] S. G. Hart and L. E. Staveland, "Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research," in *Human Mental Workload*, ser. Advances in Psychology, P. A. Hancock and N. Meshkati, Eds. Amsterdam, The Netherlands: North-Holland, 1988, vol. 52, pp. 139–183. [Online]. Available: https://doi.org/10.1016/S0166-4115(08)62386-9

[46] J. Brooke, "SUS: A 'Quick and Dirty' Usability Scale," in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[47] J.-Y. Jian, A. M. Bisantz, and C. G. Drury, "Foundations for an Empirically Determined Scale of Trust in Automated Systems," *International Journal of Cognitive Ergonomics*, vol. 4, no. 1, pp. 53–71, 2000. [Online]. Available: https://doi.org/10.1207/S15327566IJCE0401_04

[48] D. Lakens, "Calculating and Reporting Effect Sizes to Facilitate Cumulative Science: A Practical Primer for t-Tests and ANOVAs," *Frontiers in Psychology*, vol. 4, p. 863, 2013. [Online]. Available: https://doi.org/10.3389/fpsyg.2013.00863

[49] M. Farhan, M. R. Naeem, A. Almadhor, A. K. Bashir, Z. Zhu, and T. R. Gadekallu, "Explainable AI-Driven Security Framework for Cyber-Physical Production Systems in Industry 4.0: Leveraging Immersive Embedded CIoT," *IEEE Trans. Consumer Electronics*, vol. 71, no. 4, pp. 10 446–10 453, 2025. [Online]. Available: https://doi.org/10.1109/TCE.2025.3602792